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Die-cleaning attachment for workpiece transfer structure

Abstract

A die cavity cleaning element includes a plurality of air discharge openings spaced apart along the cleaning element. A shape of at least a portion of the cleaning element including the air discharge openings is structured to conform to a shape of a predetermined portion of a lower die cavity of a stamping die. The cleaning element is operable to direct a flow of pressurized air through each of the air discharge openings onto an associated part of the predetermined portion of a lower die cavity. Collectively, the pressurized airflows expel debris from the predetermined portion of a lower die cavity.

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Background/Summary

TECHNICAL FIELD

(1) The subject matter described herein relates to the cleaning of manufacturing process tooling and, more particularly, to the automated removal of stamping process debris from selected portions of a stamping die.

BACKGROUND

(2) During stamping operations, debris such as metallic filings, shavings and/or other fragments of workpiece material may detach from the workpiece and remain in the die after the finished workpiece is extracted from the lower die. This debris may accumulate in certain portions of the die cavity during repeated operation of the die and, if not removed, may be pressed into the body of a workpiece during a stamping operation. This may cause defects in part appearance and/or other quality problems. The defective stamped part must then be reworked offline or scrapped, causing delays and added expense in the production process. In addition, when such debris accumulates in certain portions of the die cavity, stamping operations must be halted while the die cavity is cleaned manually. Manual cleaning of the die cavity is time-consuming and difficult.

SUMMARY

(3) In one aspect of the embodiments described herein, a die cavity cleaning element is provided. The cleaning element includes a plurality of air discharge openings spaced apart along the cleaning

element. The cleaning element is structured so that a central axis of each air discharge opening extends perpendicular to an exterior surface of the cleaning element.

(4) In another aspect of the embodiments described herein, a system structured for performing a stamping process on a workpiece is provided. the system includes a lower die and a die cavity cleaning element. The lower die has a lower die cavity formed therein. The die cavity cleaning element includes a plurality of air discharge openings spaced apart along the cleaning element. A shape of at least a portion of the cleaning element is structured to conform to a shape of a predetermined portion of the lower die cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

(2) FIG. 1A is a schematic perspective view of a lower portion (or lower die) of a stamping die positioned at a die station, with a workpiece transfer structure in accordance with an embodiment described herein suspended above the lower die.

(3) FIG. 1B is a schematic plan view of the lower die and workpiece transfer structure of FIG. 1A.

(4) FIG. 2 is a schematic perspective view of one example of a clamping arrangement structured to enable a cleaning element to be positioned and secured in a wide range of spatial locations.

(5) FIG. 3 is a schematic cross-sectional view of a portion of a cleaning element in accordance with an embodiment described herein, showing the cleaning element in its cleaning position and in operation for cleaning debris from a portion of a die cavity.

DETAILED DESCRIPTION

(6) This disclosure teaches a cleaning element structured for removing debris from a die from cavity of a stamping die. The cleaning element may be mounted on a workpiece transfer structure adapted to move a workpiece into position on the die for a stamping operation. A shape of at least a portion of the cleaning element is structured to conform to a shape of a predetermined portion of the lower die cavity. In one or more arrangements, the predetermined portion of the lower die cavity may be a portion of the cavity which accumulates debris (such as metal filings) over repeated stamping operations. The cleaning element may include a plurality of air discharge openings spaced apart along the portion of the cleaning element that is shaped to conform to the predetermined portion of the lower die cavity. The cleaning element may be structured to discharge a pressurized airflow through each air discharge opening and onto an associated area of the predetermined portion of the lower die cavity. The conformal shape of the cleaning element facilitates thorough cleaning of the problematic portion of the die cavity, by ensuring a flow of pressurized air into each area of the problematic portion. In combination, the lower die and the cleaning element shaped to conform to a portion of a cavity of the lower die may form part of a system structured to perform a stamping process on a workpiece.

(7) In this description, uses of “front,” “forward,” and the like, and uses of “rear,” “rearward,” and the like, refer to the longitudinal directions of the vehicle. “Front,” “forward,” and the like refer to the front (fore) of the vehicle, while “rear,” “rearward,” and the like refer to the back (aft) of the vehicle. Uses of “inner,” “outer,” and the like refer to the lateral directions of the vehicle. Uses of

“above,” “upper,” “upward,” “below,” “lower,” “downward,” and the like refer to the vertical directions of the vehicle.

(8) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals may have been repeated among the different figures to indicate corresponding or analogous elements. In addition, the discussion outlines numerous specific details to provide a thorough understanding of the embodiments described herein. Those of skill in the art, however, will understand that the embodiments described herein may be practiced using various combinations of these elements.

(9) FIG. 1A is a schematic perspective view of a lower die portion (or “lower die”) **20** of a stamping die positioned at a die station. FIG. 1B is a plan view of the lower die of FIG. 1A. As used herein, the term “stamping” encompasses a variety of sheet metal forming manufacturing processes, such as punching using a machine press or stamping press, blanking, embossing, bending, flanging, coining, and other operations usable for transforming a workpiece made of metal or a similarly workable material into a final shape. As known in the pertinent art, a complementary upper die portion (or “upper die”) may incorporate punches and other surfaces operable to form a workpiece into a desired shape when the upper die presses portions of the workpiece into a cavity **22** formed in the lower die. In FIGS. 1A and 1B, the upper die of the stamping die has been omitted for clarity.

(10) One or more die cavities may be formed in the lower die **20**. Lower die **20** of FIGS. 1A and 1B has a single die cavity **22**. As known in the pertinent art, the die cavity **22** is a recess formed in the lower die **20** and into which at least a portion of the workpiece is deformed by direct or indirect contact with an upper die moving toward the lower die **20** when the workpiece is positioned over (or inside) the die cavity **22**. The upper and lower portions of the stamping die may be mounted in a suitable stamping press. With the upper and lower dies separated, a workpiece (e.g., a large sheet metal blank) (not shown) may be positioned so as to overlie (or reside in) the die cavity **22**. The upper die may then be moved into contact with the workpiece to press portions of the workpiece into the lower die cavity **22**, thereby forming the workpiece into a desired shape in a known manner.

(11) During stamping operations, debris such as filings, shavings and/or other fragments of workpiece material may detach from the workpiece and remain in the die after the finished workpiece is extracted from the lower die. This debris may accumulate in certain portions (herein referred to as “problematic portions” of the die cavity) during repeated operation of the die and, if not removed, may be pressed into the body of a workpiece during a stamping operation. This may cause defects in part appearance and/or other quality problems. The defective stamped part must then be reworked offline or scrapped, causing delays and added expense in the production process.

(12) With a view to removing stamping debris from the die cavity **22** before part quality is adversely affected, FIGS. 1A and 1B also show embodiments **24** and **26** of a die cavity cleaning element structured to be adjustably positionable to direct a pressurized airflow onto at least one predetermined problematic portion of the die cavity **22**, to remove debris from the predetermined portion of the die cavity. For simplicity, the following description will be directed primarily to cleaning element **24** as shown. It will be understood that the basic structure and operation of cleaning element **26** may be the same as the structure and operation of cleaning element **24**, except that the shape of cleaning element **26** may be different so as to conform to the shape of a different portion of the die cavity.

(13) In one or more arrangements, the cleaning element **24** may be fabricated from a hollow pipe or tube formed from a rigid material (e.g., a metal or a suitable polymer) and defining an interior **24a** (FIG. 3) structured to facilitate a flow of pressurized air therealong. The rigidity of the cleaning element material may enable the shape of the cleaning element **24** to be maintained during use, to aid in achieving and maintaining the desired spacing between the cleaning element air discharge openings and an associated portion of the die cavity to be cleaned.

(14) Referring to FIGS. 1A and 1B, a shape of at least a portion of each cleaning element may be structured to follow or conform to the shape of a predetermined portion of the die cavity. For example, FIGS. 1A and 1B show a first cleaning element **24** structured to conform to a shape of an associated predetermined first portion **22a** of the die cavity **22**, and a second cleaning element **26** structured to conform to a shape of an associated predetermined second portion **22b** of the die cavity. The first and second portions **22a**, **22b** of the die cavity **22** may be areas determined to be problematic with regard to accumulation and/or retention of debris during stamping operations.

(15) Referring to FIGS. 1A, 2, and 3, in one or more arrangements, the cleaning element **24** may also include a plurality of air discharge openings **24b** spaced apart along portions of the cleaning element. The air discharge openings **24b** may be structured to enable fluid communication between the cleaning element interior **24a** and an exterior of the cleaning element. The cleaning element **24** may be structured to be positionable in a cleaning position with respect to the die cavity **22**. When in the cleaning position, due to the overall shape of the cleaning element **24**, a distance **D1** from each air discharge opening **24b** to a predetermined surface **22p** located inside the lower die cavity **22** and along a central axis **X1** of the air discharge opening **24b** opposite the opening may be the same (within specified tolerance limits) for each air discharge opening. The predetermined die cavity surface **22p** opposite the air discharge opening(s) **24b** may correspond to the problematic portions of the die cavity **22**. In one or more arrangements, the distance **D1** may be 1 ± 0.5 inches. Thus, for example, the distance from each air discharge opening **24b** to a portion of the die cavity residing opposite the opening along a respective central axis of the opening may be 1 ± 0.5 inches. Other distances are also possible.

(16) Referring to FIG. 3, in particular embodiments, each air discharge opening **24b** may be structured so that a central axis **X1** of the opening extends perpendicular to an outer surface **24c** of the cleaning element **24** at the location of the opening **24b**. In embodiments of the cleaning element having a circular or cylindrical cross-section through a central axis **X2** of the cleaning element, the central axis **X1** of the opening **24b** may intersect the central axis **X2** of the cleaning element **24**.

(17) In one or more arrangements, the cleaning element **24** is structured so that the air discharge openings **24b** are evenly spaced apart along the cleaning element. In particular arrangements, the air discharge openings **24b** are spaced apart a distance **D2** in the range of 1.3 ± 0.3 inches.

(18) In one or more arrangements, the cleaning element **24** is structured so that the air discharge openings **24b** have equal sizes. In particular arrangements, the air discharge openings **24b** are circular with diameters of $1\text{ millimeter}\pm0.3\text{ millimeter}$.

(19) Referring to FIG. 3, to help clear debris **99** from a problematic portion of the die cavity, the cleaning element **24** may be structured so that each air discharge opening **24b** of the cleaning element directs an associated pressurized airflow **23** therethrough at a non-zero angle θ (i.e., at an “air discharge angle”) with respect to a horizontal plane **H1** extending through the opening **24b**.

(20) Referring again to FIGS. 1A and 1B, to move a workpiece to and from the die and between successive die stations and to enable adjustable positioning of the cleaning element(s) **24**, **26** with respect to the die cavity for cleaning operations, a workpiece transfer structure (generally designated **28**) may be employed. The workpiece transfer structure **28** may be structured to be movable with respect to the lower die **20** and to adjustably support the cleaning elements **24**, **26** in respective cleaning positions of the cleaning elements. The term “adjustably support” refers to an ability of the workpiece transfer structure **28** to enable air discharge angles θ of any cleaning element and spatial positions of the cleaning elements **24**, **26** with respect to the die cavity **22** to be adjusted according to the requirements of a particular application, and also to enable a height **H1** of the cleaning member above a reference surface to be adjusted as needed.

(21) In one or more arrangements, workpiece transfer structure **28** may include a cross-bar **30** structured to be movable with respect to the lower die **20**. The cross-bar **30** may be hollow to enable air supply lines, vacuum lines, electrical and/or mechanical control cables, and other elements of the workpiece transfer structure to be incorporated into the interior of the cross-bar **30**.

Alternatively, one or more of the elements (or portions of the elements) supported by the cross-bar **30** may be mounted along an exterior of the cross-bar. For example, FIGS. **1A** and **1B** show portions of air supply lines **32** for the cleaning elements **24**, **26** described herein, extending from the cross-bar **30** along the cross-bar exterior.

(22) Referring to FIGS. **1A** and **1B**, opposite ends **30a**, **30b** of the cross-bar **30** may be attached to one or more automated mechanism(s) (not shown) configured to position the workpiece at a desired location on the lower die **20** for stamping. For example, in one or more arrangements, each end of the cross-bar **30** may be operably connected to one of associated programmable robotic arms **31a**, **31b**. The term “operably connected,” as used throughout this description, can include direct or indirect connections, including connections without direct physical contact. The robotic arms **31a**, **31b** may operate in a coordinated fashion to move the cross-bar **30** and an attached workpiece in directions **S1**, **S2** with respect to the lower die **20**.

(23) Referring to FIGS. **1A** and **1B**, to support the cleaning element **24** on the cross-bar **30**, one or more support stalk(s) **34** may extend from the cross-bar **30**. In one or more arrangements, to enable positioning of cleaning elements **24**, **26** in their respective cleaning positions, the support stalk(s) **34** and any other members supporting the cleaning elements **24**, **26** may be configured (or configurable) so that the cleaning elements will automatically reside in their respective cleaning positions when the workpiece transfer structure **28** is moved to a position needed to place a blank workpiece in the die cavity **22**. Thus, in this configuration, the cleaning elements **24**, **26** will be ready to perform a cleaning operation when the workpiece transfer structure **28** (without a workpiece attached) is moved into its normal position for placement of a workpiece into the lower die cavity **22**.

(24) Referring to FIG. **3**, in other arrangements, the support stalk(s) **34** and any other members supporting the cleaning elements **24**, **26** may be configured (or configurable) so that the cleaning elements will automatically reside at locations above their respective cleaning positions when the workpiece transfer structure **28** is moved (without a workpiece attached) to a position needed to place a blank workpiece in the die cavity **22**. Thus, in these other arrangements, the cleaning elements **24**, **26** may need to be lowered vertically into the lower die cavity **22** (or deeper into the lower die cavity) to reach their respective desired cleaning positions. Thus, in certain arrangements, the cleaning elements **24**, **26** may be mounted to the workpiece transfer structure **28** so as to enable a height **H1** of the cleaning member above a reference surface to be adjusted by moving the workpiece transfer structure **28**. In particular arrangements, the reference surface may be a floor surface (not shown) of a manufacturing facility in which the die station is positioned. In other arrangements, the reference surface may be a floor or lowermost surface **R1** of the die cavity **22**. Other reference surfaces are also possible. To control the relative heights of the cleaning elements **24**, **26**, the robotic arm(s) **31a**, **31b** controlling movement of the workpiece transfer structure **28** may be programmable and operable to raise and lower the workpiece transfer structure **28** by desired amounts according to whether a cleaning operation or a workpiece placement operation is being performed.

(25) Other mechanisms may also be employed to enable “fine-tuning” adjustments to the positions of the cleaning elements, in order to achieve a desired cleaning position where the air discharge openings of the cleaning elements will expel pressurized air onto problematic portions of the die cavity. For example, with reference to FIGS. **2** and **3**, the cleaning element **24** may be mounted to the workpiece transfer structure **28** so as to be rotatable with respect to the workpiece transfer structure. In particular arrangements, the cleaning element **24** may be mounted on the workpiece transfer structure **28** so as to enable an air discharge angle θ of at least one air discharge opening **24b** to be adjusted.

(26) FIG. **2** shows one example of a clamping arrangement structured to enable the cleaning element **24** to be positioned and secured in a wide range of spatial locations. In some embodiments, the clamping arrangement may include a clamp **40a** adjustably connecting a cleaning element

support stalk **34** with a stalk extension **37** to enable translational adjustment of the cleaning element **24** along an axis **C1** and rotational adjustment of the cleaning element **24** about axis **C1** in directions **T1** and **T2**.

(27) In some embodiments, the clamping arrangement may also include a dual clamp **40b** adjustably connecting the stalk extension **37** with the cleaning element **24**. The dual clamp **40b** may enable translational adjustment of the cleaning element **24** along an axis **C2** and rotational adjustment of the cleaning element **24** about axis **C2** in directions **T3** and **T4**. In some arrangements, axis **C2** may be perpendicular to axis **C1**. The dual clamp may also enable translational adjustment of the cleaning element **24** along another axis **C3** and rotational adjustment of the cleaning element **24** about axis **C3** in directions **T5** and **T6**. In some arrangements, axis **C3** may be perpendicular to axis **C2**. Other clamping arrangements are also possible.

(28) The flexibility provided by the possible clamping arrangements may enable fine-tuning of the position of the cleaning element **24** when mounted on the cross-bar **30**, to help achieve a desired optimum cleaning position of the cleaning element. Mechanisms other than clamps may also be employed to enable “fine-tuning” adjustments to the positions of the cleaning elements.

(29) All of the cleaning elements mounted on the workpiece transfer structure **28** may be connected in parallel with the pressurized air supply controlled by one or more valves so that air can be supplied to all cleaning elements simultaneously. By this system, all problematic portions of the lower die **20** may be cleaned simultaneously during a cleaning operation.

(30) Referring again to FIGS. **1A** and **1B**, for transporting a workpiece between die stations (or to and from a die station), the workpiece transfer structure **28** may include suction element(s) **44** structured to be operable to releasably secure the workpiece to the workpiece transfer structure so as to enable movement of the workpiece by moving the cross-bar **30**. The suction elements **44** may include associated suction cups **44a**, with each cup being structured to maintain a vacuum between the cup and the workpiece in a known manner, thereby securing a portion of the workpiece to the cup. For purposes of applying a vacuum to each cup **44a** after the cup has been brought into contact with the workpiece, each suction element **44** may include an associated hollow attachment stalk **44b** in gas-tight fluid communication with a surface of the cup facing the workpiece. The hollow portion of the stalk **44b** may also be in gas-tight fluid communication with vacuum tubing or piping (not shown) fluidly communicating with an associated vacuum generation mechanism (not shown). The suction element(s) **44** are structured to releasably secure the workpiece to the workpiece transfer structure in that, when the suction element(s) **44** are brought into contact with suitable surfaces of the workpiece and a vacuum is applied to the suction element(s) **44**, the suction element(s) **44** will attach or secure to the workpiece in a known manner. When the vacuum is removed from the suction element(s) **44**, the workpiece will be released from attachment to the suction element(s) **44**.

(31) The positions and configurations of the suction elements **44** may be specified according to the geometry of each particular workpiece, to conform to the portions of the workpiece suitable for adhering to the vacuum cups **44a**. For example, the configurations of the attachment stalks **44b** and the locations on the cross-bar **30** from which the attachment stalks **44b** extend may be tailored so that the vacuum cups **44a** contact relatively flat, gently-curved and/or other surfaces of the workpiece amenable to vacuum attachment, to enable the workpiece to be securely “gripped” by the workpiece transfer assembly for transport.

(32) In addition, the configurations of the cleaning element support stalks **34** (and other elements support the cleaning elements) may be tailored to the geometries of the suction cup attachment stalks **44b**, to enable the cleaning element support stalks **34** to remain mounted on the cross-bar **30** along with the suction element attachment stalks **44b**. This enables a cleaning operation to be performed at any desired time, by positioning the workpiece transfer structure **28** over the die cavity **22** without a workpiece attached so that the cleaning elements **24**, **26** reside in their respective cleaning positions. Pressurized air may then be directed through the cleaning elements

24, 26 to clean the die cavity **22**.

(33) All of the suction elements **44** may be connected in parallel with the suction controlled by one or more valves so that a vacuum can be applied and maintained in all suction elements simultaneously to secure the workpiece to the workpiece transfer structure **28**. By this system, air may also be admitted to all of the suction elements **44** at once, thereby eliminating the localized vacuums and releasing the workpiece from the workpiece transfer structure **28**.

(34) Referring again to FIG. **3**, during a cleaning operation, a valve may be operated to enable pressurized air to flow along the cross-bar **30**, through air supply line(s) **32**, then into the cleaning elements **24, 26**. The pressurized air is then expelled through the cleaning element air discharge openings and onto the problematic portions of the die cavity **22** to expel debris from the die cavity. The air supply may then be stopped, and stamping operations may be resumed by operating the suction elements **44** to affix another workpiece to the workpiece transfer structure **28**.

(35) As described herein, each of die cavity cleaning elements **24, 26** may be custom-designed to conform to the shape of a particular portion of the die cavity **22**, and to include the characteristics just described. The overall shape of a cleaning element, the distribution of the air discharge openings along the cleaning element, and the spacing of the air discharge openings from the problematic areas of the die cavity may help evenly distribute the pressurized airflow to a given problematic area and ensure that a problematic area of the die cavity receives pressurized air streams sufficient to eject accumulated debris.

(36) Optimum discharge angles θ of the air discharge openings and cleaning position parameters such as optimum distances of the air discharge openings from particular problematic surfaces of a given die cavity may be determined iteratively, through experimentation. A suitable cleaning element may then be tailored specifically for a given set of particular problematic surfaces or problematic portion of the die cavity.

(37) In addition, die cavity cleaning operations may be automated and incorporated into normal stamping operations as needed. For example, if it is determined that debris must be removed from portions of the die cavity at least every 30 cycles of the stamping die, automated operation of the workpiece transfer structure may be programmed to bypass attachment of the transfer structure to a new workpiece for processing and, instead, to move the cross-bar (without a workpiece attached) so as to place the cleaning element(s) in their respective cleaning positions over (or inside) the die cavity. Pressurized air may then be directed through the air discharge openings to expel debris from the portion(s) of the die cavity conforming to the shape(s) of the cleaning element(s).

(38) Operation of the workpiece transfer structure may also be programmed to automatically resume the process flow after the cleaning operation, by automatically releasably securing a new workpiece to the transfer structure and moving the workpiece into position on the lower die for stamping operation. In this manner, any required cleaning operations may be automated and incorporated into the process flow, without interruption or human involvement.

(39) Also, by using multiple cleaning elements mounted on the workpiece transfer structure and with each cleaning element conforming to an associated problematic portion of the die cavity, the structures described herein enable each problematic portion of the die cavity to be cleaned simultaneously with all other problematic portions of the die cavity.

(40) Detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in FIGS. **1A-3**, but the embodiments are not limited to the illustrated structure or application.

(41) The terms “a” and “an,” as used herein, are defined as one or more than one. The term

“plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . .” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC or ABC).

(42) Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

Claims

1. A workpiece transfer structure structured for positioning a workpiece in a stamping die, the workpiece transfer structure comprising: a cross-bar structured to be movable with respect to the stamping die; at least one suction element structured to be operable to releasably secure the workpiece to the cross-bar so as to enable movement of the workpiece by moving the cross-bar; and a die cavity cleaning element including a plurality of air discharge openings spaced apart along the cleaning element, the cleaning element being structured so that a central axis of each air discharge opening extends perpendicular to an exterior surface of the cleaning element.
2. The cleaning element of claim 1, structured so that a central axis of each air discharge opening intersects a central axis of the cleaning element.
3. The cleaning element of claim 1, structured so that the openings of the plurality of air discharge openings are evenly spaced apart along the cleaning element.
4. The cleaning element of claim 1, wherein a shape of at least a portion of the cleaning element is structured to conform to a shape of a predetermined portion of an associated die cavity of a stamping die.
5. The workpiece transfer structure of claim 1, further comprising: a support stalk extending from the cross-bar; and at least one clamp mounted on the support stalk and structured to adjustably support the cleaning element on the support stalk.
6. A system structured for performing a stamping process on a workpiece, the system comprising: a lower die having a lower die cavity formed therein; and a die cavity cleaning element including a plurality of air discharge openings spaced apart along the cleaning element, the cleaning element being positionable in a cleaning position with respect to the die cavity, in which a distance from each air discharge opening to a predetermined surface located inside the lower die cavity and along an axis of the air discharge opening is equal, a shape of at least a portion of the cleaning element being structured to conform to a shape of a predetermined portion of the lower die cavity; and a workpiece transfer structure structured to be movable with respect to the lower die and to adjustably support the cleaning element in the cleaning position.
7. The system of claim 6, wherein the cleaning element is structured so that a central axis of each air discharge opening extends perpendicular to an exterior surface of the cleaning element.
8. The system of claim 6, wherein the cleaning element is structured so that a central axis of each air discharge opening intersects a central axis of the cleaning element.
9. The system of claim 6, wherein the cleaning element is structured so that the openings of the plurality of air discharge openings are evenly spaced apart along the cleaning element.
10. The system of claim 6, wherein the cleaning element is mounted on the workpiece transfer structure so as to enable an air discharge angle of at least one air discharge opening of the cleaning element to be adjusted.
11. The system of claim 10, wherein the workpiece transfer structure includes: a cross-bar structured to be movable with respect to the lower die; a support stalk extending from the cross-bar; and at least one clamp mounted on the support stalk and structured to adjustably support the

cleaning element on the support stalk.

12. The system of claim 10, wherein the workpiece transfer structure includes at least one suction element structured to be operable to releasably secure the workpiece to the workpiece transfer structure so as to enable movement of the workpiece by moving the workpiece transfer structure.
 13. The system of claim 10, wherein the cleaning element is mounted to the workpiece transfer structure so as to enable a height of the cleaning element above a reference surface to be adjusted by moving the workpiece transfer structure.
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